

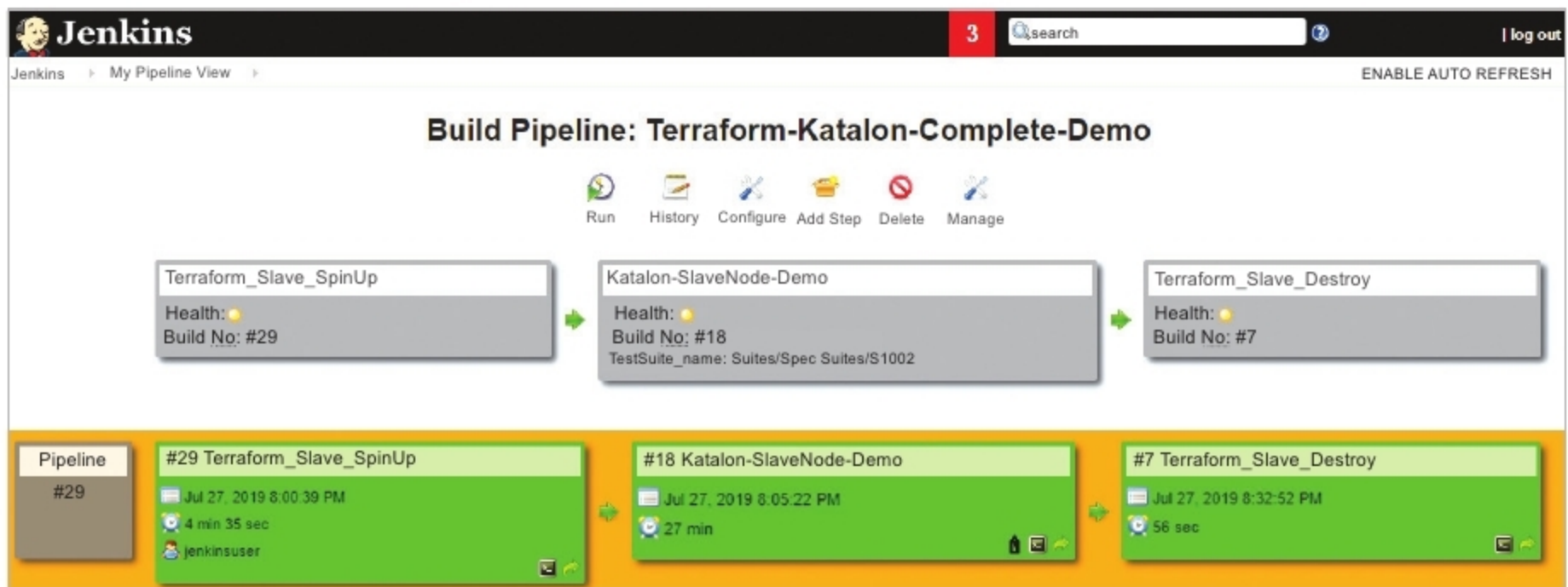


Figure 1: Jenkins build pipeline

CI tool (Jenkins): Jenkins is an open source, self-reliant automation server initially developed for the implementation of continuous integration (CI), a development practice. From Jenkins 2 onwards, this server has seen exceptional growth, providing hundreds of plugins to support building, deployment and automation in any project. It also has a Terraform plugin for Jenkins, which allows for the creation of test infrastructure before running the build, and its deletion after the build is completed.

Infrastructure-as-a-Code approach: Terraform is a cloud-agnostic tool for building, changing and versioning infrastructure safely and efficiently. It can manage existing and popular cloud service providers like AWS, Azure and GCP, as well as custom in-house solutions. It helps in modelling the complete cloud Infrastructure-as-a-Code, i.e., the infrastructure is described using a high-level configuration syntax. This allows a blueprint of the infrastructure to be versioned and treated as you would any other code. Additionally, infrastructure can be shared and reused. This way the time required to manage the resources is reduced. There is no need to individually create and configure the cloud resources and figure out the

relevant interdependencies. Terraform handles all of that.

This Terraform template will launch an EC2 instance running a custom Amazon Machine Image (AMI) in a public subnet within a virtual private cloud (VPC) with an attached Internet gateway. The route tables associated with the public subnet can be configured to be accessed only from the organisation's public IP as a security precaution. The role required for the instance to upload data to the S3 bucket

can also be defined within the template resources. The custom Amazon AMI has all the required software installed and configured to run the test suite and initiate the test execution. This saves setup time. Once the execution is complete the reports can be uploaded to an S3 bucket by placing the request via AWS CLI (command line interface).

Version control: Git is an open source distributed version control system. The latest code for the Katalon test suites is maintained in Git; this

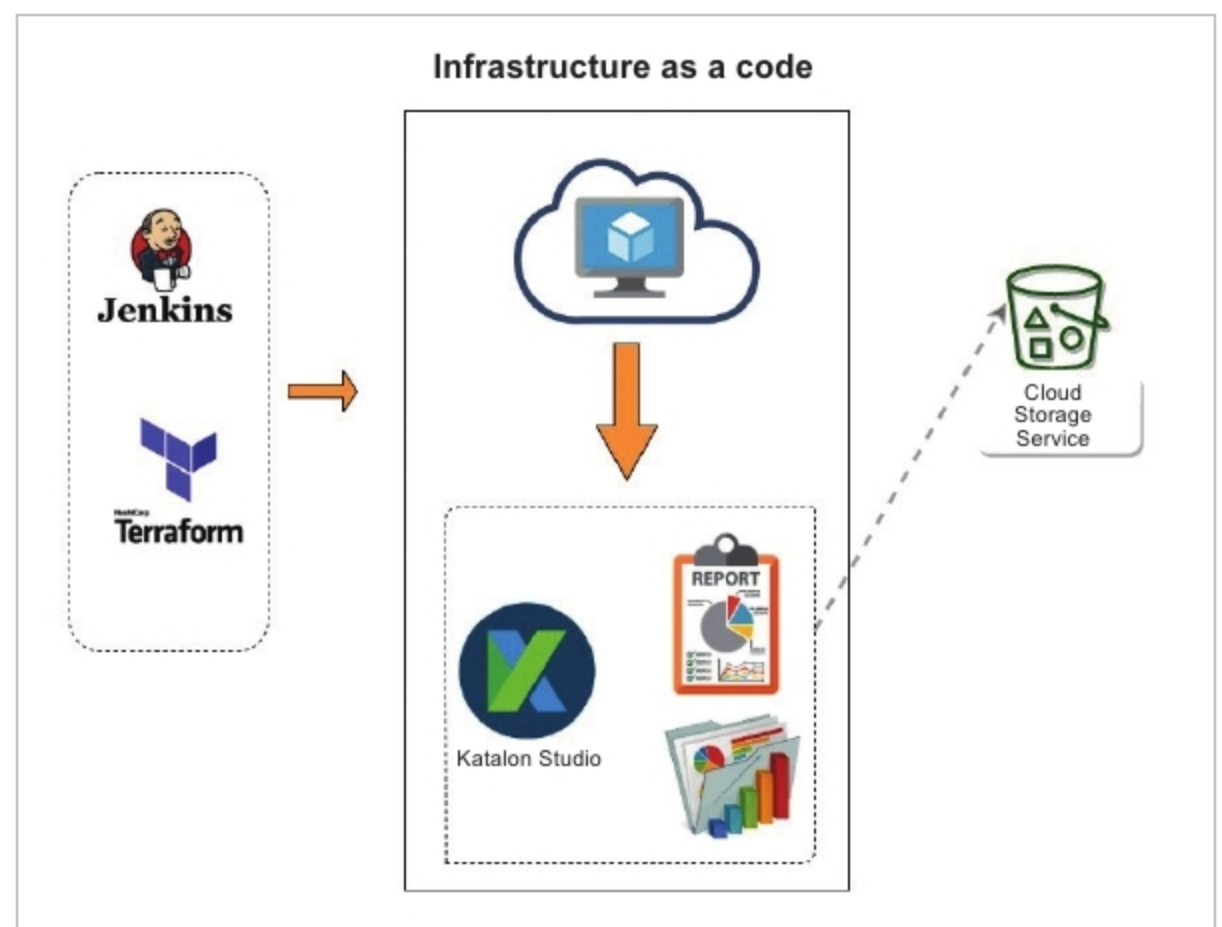


Figure 2: Proposed architecture design